

Arithmophysics: A Direct Subpower Closure Attempt Through the Horizon Map

Research directive for Christophe Michaels's light-plane programme.

Read the supplied papers, Michaels_Light_Plane_Horizon_Map(1).pdf and Michaels_Horizon_Map_Audit(1).pdf, including the audit's follow-up on phase readout and omitted tails. Use their exact definitions and retained terminal contributions. This prompt also supplies two elementary extensions: a converse mode-population criterion and the exact prime action in the horizon-mode basis.

The objective is a direct analytical closing attempt. Determine whether the arithmetic source's prime structure forces the subpower estimate below. Carry the calculation through the explicit horizon map, identify the mechanism that controls amplified modes, and derive the all-scale bound if that mechanism is sufficient. Execute the investigation rather than returning another research plan.

The larger-space viewpoint is the working method: calculate the geometry and arithmetic together, then carry the result back through the projection. Preserve the distinction between a governing law used for exploring the model and a law derived from independent inputs. In the closing derivation, the subpower law (G), its equivalent mode bound, and any consequence used to recover them must have an independent justification. State this dependency distinction once, then concentrate on the calculations.

The desired outcome is a complete proof with constants and quantifiers. If an attempted mechanism falls short, preserve every proved identity and return the smallest explicit remaining estimate together with the work done toward it.

Use the following exact objects.

Let

$$M(x) = \sum_{n \leq x} \mu(n), \quad R(N) = \sum_{a, b \leq N} \frac{\mu(a)\mu(b)}{\max(a, b)} = \sum_{k < N} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}.$$

The terminal term is part of the target. The complete field is

$$G_v(u) = e^{-|u-v|/2}, \quad H_0 = -\partial_u^2 + \frac{1}{4}, \quad \phi_N = \sum_{n \leq N} \frac{\mu(n)}{\sqrt{n}} G_{\log n},$$

$$\|f\|_E^2 = \int_{\mathbb{R}} (|f'|^2 + \frac{1}{4}|f|^2) du, \quad \|\phi_N\|_E^2 = R(N).$$

On the source space $\ell^2(\mathbb{N})$, let

$$Z = 15/\pi^2, \quad \Psi_n = \frac{\mu(n)}{n\sqrt{Z}}, \quad H_{\text{src}}|n\rangle = (\log n)|n\rangle,$$

$$P_N|n\rangle = \sqrt{Zn} G_{\log n} \quad (n \leq N), \quad P_N|n\rangle = 0 \quad (n > N).$$

Then $P_N\Psi = \phi_N$, and the retained matrix is

$$P_N^* P_N = Z B_N, \quad (B_N)_{ab} = \min(a, b).$$

The growth target concerns the actual source and this observable. Source norm conservation and the observable's largest eigenvalue alone do not quantify their alignment.

For $1 \leq j, n \leq N$, define

$$\theta_{j,N} = \frac{(2j-1)\pi}{2N+1}, \quad v_{j,N}(n) = \frac{2}{\sqrt{2N+1}} \sin(n\theta_{j,N}), \quad \lambda_{j,N} = \frac{1}{4 \sin^2(\theta_{j,N}/2)}.$$

Let V_N have columns $v_{j,N}$, let $\Lambda_N = \text{diag}(\lambda_{1,N}, \dots, \lambda_{N,N})$, and set

$$c_N(n) = \frac{\mu(n)}{n}, \quad b_N = V_N^* c_N, \quad S_N(J) = \sum_{j \leq J} |b_{j,N}|^2.$$

The exact resolution is

$$R(N) = \sum_{j=1}^N \lambda_{j,N} |b_{j,N}|^2.$$

The gains decrease with j , and uniformly in $1 \leq j \leq N$ they are comparable, by absolute constants, to N^2/j^2 .

Target the following equivalent criterion.

Define

$$K_N = \max_{1 \leq J \leq N} \lambda_{J,N} S_N(J).$$

For every finite source, not just the Möbius source,

$$K_N \leq R(N) \leq \left(1 + \log \frac{\lambda_{1,N}}{\lambda_{N,N}}\right) K_N. \quad (\text{A})$$

To verify the upper bound, use

$$R(N) = \lambda_{N,N} S_N(N) + \sum_{j < N} (\lambda_{j,N} - \lambda_{j+1,N}) S_N(j)$$

and $S_N(j) \leq K_N / \lambda_{j,N}$, followed by $1 - r \leq -\log r$ for $0 < r \leq 1$. The lower bound follows from decreasing gains. The multiplier in (A) is $O(1 + \log N)$.

Thus the closing target can be written

$$\boxed{\forall \varepsilon > 0 \exists C_\varepsilon < \infty \forall N \geq 1 : K_N \leq C_\varepsilon N^\varepsilon.} \quad (\text{T})$$

Equivalently, with constants independent of both N and J ,

$$S_N(J) \leq C_\varepsilon J^2 N^{-2+\varepsilon} \quad (1 \leq J \leq N). \quad (\text{T}')$$

Either target gives $R(N) = O_\varepsilon(N^\varepsilon)$ by allocating a smaller exponent before absorbing the logarithmic loss. Conversely, subpower $R(N)$ implies these targets.

The return to arithmetic is explicit: $M(N)^2/N \leq R(N)$. The classical criterion $M(x) = O_\delta(x^{1/2+\delta})$ for every $\delta > 0$ is equivalent to RH. Conversely, inserting that criterion into the historical-plus-terminal identity makes $R(N)$ subpower. An accessible primary reference for the classical criterion is Lynelle Ye, Bounding sums of the Möbius function over arithmetic progressions, <https://arxiv.org/abs/1406.7326>.

This is the precise sense in which (T) is a complete closing target. A bound on a fixed number of modes, a finite set of horizons, or one fixed positive exponent has a narrower quantified conclusion.

Begin the arithmetic investigation with this exact prime action.

For a finite prime set Q , define $\mu_Q(n) = \mu(n)$ if every prime divisor of n lies in Q , and zero otherwise; set $\mu_Q(1) = 1$. Write

$$c_{Q,N}(n) = \mu_Q(n)/n, \quad b_{Q,N} = V_N^* c_{Q,N}.$$

For $p \notin Q$, $p \leq N$, and $K = \lfloor N/p \rfloor$, let $E_{p;N,K} : \mathbb{C}^K \rightarrow \mathbb{C}^N$ be the embedding

$$E_{p;N,K} e_m = e_{pm}.$$

Define the explicitly computable transfer matrix

$$T_{p;N,K} = V_N^* E_{p;N,K} V_K, \quad (T_{p;N,K})_{j\ell} = \sum_{m=1}^K v_{j,N}(pm) v_{\ell,K}(m).$$

Then

$$\boxed{b_{Q \cup \{p\},N} = b_{Q,N} - p^{-1} T_{p;N,K} b_{Q,K}.} \quad (\text{B})$$

This follows coefficient by coefficient from $c_{Q \cup \{p\},N} = c_{Q,N} - p^{-1} E_{p;N,K} c_{Q,K}$. For $p > N$, the correction is empty.

Two identities fix the normalization and metric:

$$T_{p;N,K}^* T_{p;N,K} = I_K, \quad T_{p;N,K}^* \Lambda_N T_{p;N,K} = p \Lambda_K. \quad (\text{C})$$

The second follows from $E_{p;N,K}^* B_N E_{p;N,K} = p B_K$. Accordingly, the squared energy of the correction in (B) is $p^{-1} R_Q(K)$, in agreement with the exact field update

$$\phi_{Q \cup \{p\},N} = \phi_{Q,N} - p^{-1/2} \mathcal{T}_{\log p} \phi_{Q,K}, \quad (\mathcal{T}_a f)(u) = f(u - a).$$

Let $\Pi_{J,N}$ retain the first J mode coordinates. The exact cumulative-population update is

$$\begin{aligned} S_{Q \cup \{p\},N}(J) &= S_{Q,N}(J) + p^{-2} \|\Pi_{J,N} T_{p;N,K} b_{Q,K}\|^2 \\ &\quad - \frac{2}{p} \operatorname{Re} \langle \Pi_{J,N} b_{Q,N}, \Pi_{J,N} T_{p;N,K} b_{Q,K} \rangle. \end{aligned} \quad (\text{D})$$

Retain this signed interaction. The amplitudes at horizons N and K come from the same arithmetic source.

The initial source is $c_{\emptyset,N} = e_1$. Taking all primes through N gives the actual full Möbius source at that horizon. If an induction uses intermediate prime sets, specify the admissible sets and any dependence of its bounds on the retained primes. A claim for the final arithmetic source must not silently become a claim for arbitrary vectors or every partial prime set.

Carry out the following proof attempt.

1. Verify (A)–(D) analytically and resolve any indexing or conjugation issue before using them. Keep the exact cutoff $\lfloor N/p \rfloor$, terminal row, and mixed terms.
2. Evaluate the finite sine-product sum defining $T_{p;N,K}$. Obtain formulas and bounds for transfer into the first J modes, including the resonant indices, aliases caused by sampling at multiples of p , and floor-dependent errors. State their uniform ranges in N, p, J ; treat vanishing denominators by their exact limits.
3. Use (D) to investigate cancellation in complete prime packets. Combine correlated contributions before taking absolute values. Search for an identity, inequality, or conserved quantity tying the cross term to the two squared terms across the linked horizons. Quantify what actual Möbius multiplicativity contributes beyond the identities valid for every source.
4. Attempt to prove (T) or (T') directly. Accept an equivalent sufficient estimate if its transfer to the complete $R(N)$ is proved. A useful alternative is a uniform subpower bound for the energy in each dyadic mode band: there are only $O(1 + \log N)$ such bands. Parseval gives $\sum_j |b_{j,N}|^2 = \sum_{n \leq N} \mu(n)^2 / n^2 \leq Z$; exploit this where it already controls high-index modes.

5. If a recursive or invariant-region argument emerges, carry it through to arbitrary N . Display the initial data, every accumulated error, the dependence on ε , and the treatment of horizons between any selected scales. Demonstrate that the resulting exponent can be made arbitrarily small. If a route achieves only a fixed exponent, identify the exact term that sets it and try a signed packet estimate at that term.
6. Use the common two-cutoff field from the audit when comparing prime and integer truncations. Track the products omitted by the integer horizon explicitly. A bound for a full prime product must include a justified passage to each integer prefix before it can be used in (T).
7. Use computation to choose or reject concrete analytic mechanisms: locate the J attaining K_N , measure which prime packets change it, and compare the exact cross term with any proposed bound. Attach those computations to stated inequalities and reproducible inputs. An all-scale conclusion must be supported by an all-scale argument.

The uniform phase-transport theorem is available when useful:

$$q(\tau)^{-2}R(N) \leq R_N(\tau) \leq q(\tau)^2R(N), \quad q(\tau) = \sqrt{1 + \tau^2} + |\tau|.$$

Use it to move between readouts after identifying a useful phase estimate. A long-phase mean or a finite observed correlation sign needs its own quantified comparison before it supplies an upper bound for the target readout.

Return a concrete research result.

Lead with the strongest theorem actually obtained and its inputs. If (T) closes, provide the full argument through (A), the terminal bound, and the classical Möbius criterion. Include the exact invariant or cancellation mechanism responsible.

If the closing estimate is still unavailable, return the proved transfer formulas and bounds, followed by one explicit residual inequality. Specify its source class, constants, quantifiers, sign requirements, and dependencies. Show where it enters the closing chain, why the attempted estimates stop there, and the strongest refinement actually tested or proved. This residual should be sharper or more informative than simply repeating (T).

Keep the effort directed at closure. Every auxiliary investigation should either prove a needed estimate, expose a concrete failure of a proposed mechanism, or sharpen the remaining arithmetic comparison.